

Pragya Sharma

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Summary: Computer Engineering Ph.D. candidate (graduating in Dec 2026) with research focus in applied AI/ML for telecom networks and network security. Experienced in developing end-to-end distributed software systems and AI/ML workflows for data-driven applications. Hands-on experience building domain-specific LLM systems using hybrid RAG, LoRA fine-tuning, and systematic evaluation. Prior full-time software engineering experience of 2.5 years in DSP systems.

EDUCATION

Virginia Tech

Ph.D. in Electrical and Computer Engineering (ECE) (GPA: 4.0/4.0)

Alexandria, VA, USA

Jan 2021 – Dec 2026 (Expected)

Indian Institute of Technology Bombay

B.Tech. & M.Tech. in Electrical Engineering (GPA: 8.36/10.0)

Mumbai, India

Jul 2013 – Jun 2018

WORK EXPERIENCE

Nokia Bell Labs

AI-based Cybersecurity Research Intern

Murray Hill, NJ, USA

Jun 2026 – Aug 2026

- Built an automated 5G user-plane security testbed (Open5GS core + UERANSIM), orchestrating controlled degradation experiments that inject on-path availability attacks (packet drop, rate throttling, delay, jitter).
- Engineered a Python/Bash data pipeline that derives transport-layer KPIs from raw PCAPs, producing labeled TCP/UDP KPI datasets for downstream causal modeling.
- Developed a causal attribution engine using structural causal models (SCMs) to quantify how each attack impacts network telemetry, validating every estimate against independent refutation tests.

Futurewei Technologies

Wireless Research Intern

Addison, TX, USA

May 2025 – Aug 2025

- Designed 6G core network architecture enhancements, enabling QoS-aware beyond-connectivity service sharing (e.g., sensing) between local subnetworks of industry verticals.
- Developed Python-based network simulations and analyzed 3GPP SA2 technical reports and specifications to propose a standards-aligned solution.

A2 Labs (formerly Kryptowire Labs)

Research & Development Intern

Arlington, VA, USA

May 2022 – Aug 2022

- Designed and integrated a decentralized service authorization framework for 5G core network functions, deploying WAVE sidecar proxies alongside OpenAirInterface (OAI) NFs in an end-to-end Kubernetes-based 5G core.
- Evaluated the scalability of WAVE through stress-testing experiments with multiple distributed client/server pods on Kubernetes, characterizing authorization performance under increasing service-request workloads.
- Benchmarked end-to-end authorization latency and overhead across 5G service requests to quantify performance–security tradeoffs and validate WAVE as an authorization mechanism for 5G core networks.

Cadence Design Systems

Design Engineer II

Pune, India

Jul 2018 – Dec 2020

- Developed C-based API frameworks and benchmarking suites to evaluate Tensilica ConnX Baseband DSP cores across multiple core configurations; conducted systematic cross-compiler performance validation and analysis.
- Optimized DSP core performance through in-depth analysis of the Instruction Set Architecture, leveraging specialized instructions and architectural features to reduce cycle counts and improve overall execution efficiency.

TECHNICAL SKILLS

Programming: C, C++, SQL, Python, Bash (Linux)

ML & Data: PyTorch, scikit-learn, XGBoost, pandas, NumPy, SHAP, DoWhy, Gymnasium

LLMs: Hugging Face Transformers, PEFT, LangChain, RAG, SFT, LoRA/QLoRA, FAISS, BM25, Hybrid Search

ML Methods: Adversarial ML, Causal Inference, Federated Learning, Deep Learning, Reinforcement Learning

Networks & Telecom: OpenAirInterface (OAI), Open5GS, O-RAN, Wireshark, MATLAB, tcpdump, iperf, tc-netem

Systems & Infrastructure: Kubernetes, Docker, Git

AI Coding Platforms: Cursor, Claude Code, Codex

PUBLICATIONS

1. **P. Sharma**, S. Sun, S. Deshpande, A. Stavrou, H. Wang. “Towards xApp Conflict Evaluation with Explainable Machine Learning and Causal Inference in O-RAN”. *Published in IEEE NFV-SDN 2025 (Best Paper Award)*.
2. **P. Sharma**, T. Atalay, H. Gibbs, D. Stojadinovic, A. Stavrou, H. Wang. “5G-WAVE: A Core Network Framework with Decentralized Authorization for Network Slices”. *Published in IEEE INFOCOM 2024 (Core A* Rank)*.
3. S. Sun, **P. Sharma**, S. Maitra, A. Stavrou, H. Wang. “VulLens: Enhancing Software Vulnerability Detection against Evasion Attacks”. *Published in IEEE/IFIP DSN 2026 (Core A Rank)*.
4. S. Sun, **P. Sharma**, K. Nwodo, A. Stavrou, H. Wang. “FedMADE: Robust Federated Learning for Intrusion Detection in IoT Networks Using a Dynamic Aggregation Method”. *Published in Information Security Conference (ISC) 2024*.
5. **P. Sharma**, S. Sun, A. Stavrou, H. Wang. “Silencing the Alarm: Evasion Attacks against O-RAN Anomaly Detection xApp with Feasible Adversarial Examples”. *Under Review*.
6. **P. Sharma**, A. Xiang, A. Kiani, J. Kaippallimalil, T. Saboorian, H. Wang. “Beyond Connectivity, Beyond Boundaries: Enabling Service Sharing in Interconnected 6G Subnetworks”. *Under Review*.

PROJECTS

Telecom LLM Assistant: Hybrid RAG and LoRA Fine-Tuning

- Built a retrieval-augmented generation (RAG) pipeline over 3GPP specifications using LangChain’s recursive chunking and hybrid retrieval with BGE embeddings, FAISS, BM25, and reciprocal rank fusion (RRF).
- Fine-tuned four 3–4B open-weight LLMs—Qwen3, Llama 3.2, Gemma 3, and Phi-4-mini—on telecom QA data using supervised fine-tuning (SFT) with PEFT methods like LoRA, QLoRA.
- Evaluated base, fine-tuned, RAG, and fine-tuned-plus-RAG models on answer correctness and latency.

Adversarial Robustness of O-RAN xApps through Feasible Adversarial Samples

- Investigated the security posture of O-RAN xApps against adversarial ML attacks conducted by malicious xApps.
- Constructed feasible adversarial samples through a novel constrained PGD (C-PGD) attack algorithm to test the robustness of anomaly detection xApp against white-box evasion attacks.
- Achieved higher Attack Success Rates (ASRs) than state-of-the-art adversarial ML attacks across multiple perturbation budgets while respecting domain constraints.

xApp Conflict Evaluation in O-RAN using XAI and Causal ML

- Designed a hybrid explainable AI (XAI) and causal inference pipeline to quantify the impact of xApp control actions on network KPIs and identify potential xApp conflicts.
- Trained regression models (XGBoost, Random Forest, MLP, SVR) on network simulation data and utilized SHAP to identify influential feature–KPI relationships for constructing causal DAGs.
- Estimated Average and Conditional Average Treatment Effects (ATE/CATE) over the resulting causal models to quantify population- and context-specific impacts of xApp actions on network performance.

PolicyLens | Microsoft x CCI Innovation Challenge Hackathon 2026

- Built an evidence-grounded GenAI application using Azure OpenAI (GPT-4.1), and RAG to analyze regulatory documents, public feedback, and media coverage, producing source-cited insights with confidence labels and scores.
- Developed an LLM-based NLP analysis and fact-checking pipeline to ingest news articles, extract claims and stakeholder viewpoints, retrieve evidence (BM25), and perform claim-level verification against policy documents.

Robust Federated Learning for Intrusion Detection in IoT Networks

- Co-designed a dynamic aggregation method for federated learning that assigns adaptive weights to local client models, addressing data and device heterogeneity across distributed IoT nodes.
- Improved detection of minority attack classes in a federated intrusion detection system (IDS), boosting classification performance on imbalanced network traffic data.

AWARDS AND ACHIEVEMENTS

- **Pratt Fellowship (3x recipient)**, Dept. of ECE, Virginia Tech, 2024 & 2026.
- **Best Paper Award**, IEEE NFV-SDN, 2025.
- **Student Travel Grant to attend IEEE INFOCOM**, US Army Research Office, 2024.
- **All India Rank (AIR) 192** (top 0.1%), IIT-JEE Advanced Exam, 2013.